

## ALTIRIG · REQUIREMENTS

# AltiRig requirements

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## DOCUMENT CONTROL

|             |                                       |         |                |
|-------------|---------------------------------------|---------|----------------|
| Document ID | ALR-REQ-001                           | Version | 0.3            |
| Status      | <b>Draft</b>                          | Date    | 2026-10-03     |
| Author      | Amish Chadha                          | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept         |         |                |
| Repository  | github.com/BoujeeEnjinia1701/altirig  |         |                |
| Source      | docs/03-requirements.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                |
|---------|------------|--------------|-----------------------------------------------------------------------------------------------------------------------|
| 0.3     | 2026-10-03 | Amish Chadha | TRL 3 status from the calculations (ALR-CAL-001) and the constructable design (ALR-DDR-002)                           |
| 0.2     | 2026-10-03 | Amish Chadha | TRL 2 review; concept status for every requirement; R11 interlocks and R12 over-vacuum protection added (ALR-DDR-001) |
| 0.1     | 2026-09-30 | Amish Chadha | Initial scaffold                                                                                                      |

# AltiRig requirements

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Ten of the twelve requirements are met on paper by the constructable design; R6 (chamber effect) is at risk and R10 (cost) is not met. Targets are unchanged from the scaffold except for the two safety requirements added at the TRL 2 review (R11 and R12). Status is on paper only: every requirement is verified by test at TRL 4.

*Table 1. Requirements and their status at TRL 3.*

| ID  | Requirement               | Target                                                                                                                                                                                       | Verification (TRL 4 or later)                                                | Status at TRL 3                                                                                                                                   |
|-----|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| R1  | Density range             | Hold any air density from ambient down to 0.74 kg/m <sup>3</sup> (5,000 m equivalent) at room temperature                                                                                    | Pressure and temperature log over a 10 min hold at several set points        | Met on paper: 0.736 kg/m <sup>3</sup> needs 61.9 kPa at 20 °C; the pump reaches far lower and the relief valve stops at 46.3 kPa (ALR-CAL-001, A) |
| R2  | Density control           | Density held within plus or minus 1% of set point during a run                                                                                                                               | Logged density during motor runs                                             | Met on paper: error budget 0.6 % (ALR-CAL-001, D)                                                                                                 |
| R3  | Thrust range and accuracy | 0 to 50 N thrust with error no more than 1% of full scale                                                                                                                                    | Calibration with certified masses before and after a test series             | Met on paper: flexure stand with a 10 kg cell, error budget 0.13 N, 0.27 % (ALR-CAL-001, E)                                                       |
| R4  | Torque and speed          | Torque 0 to 2 N m within 2%; speed within 0.5%                                                                                                                                               | Calibration arm with masses; optical tachometer cross-check                  | Met on paper: torque error 0.3 %, speed 0.02 % (ALR-CAL-001, E)                                                                                   |
| R5  | Propeller size            | Propellers up to 380 mm (15 in) diameter                                                                                                                                                     | Fit check and clearance measurement                                          | Met on paper: 210 mm from tip to wall, 80 mm to the deck (ALR-CAL-001, E)                                                                         |
| R6  | Chamber effect            | At ambient pressure, chamber thrust within 5% of an open-air stand for the same propeller and speed                                                                                          | Side-by-side comparison tests                                                | At risk: estimated 2.9 % to 5.3 % high with the 63 % open baffle (ALR-CAL-001, F); see the review note                                            |
| R7  | Vessel safety factor      | Collapse pressure at least 4 times the maximum working pressure difference (target)                                                                                                          | Calculation reviewed by a qualified engineer, then proof test behind a guard | Met on paper, at full vacuum: shell 6.7, heads 40, window 7.1 on yield (ALR-CAL-001, B)                                                           |
| R8  | Pump-down time            | Ambient to 54 kPa in no more than 5 min                                                                                                                                                      | Timed pump-down with pressure log                                            | Met on paper: 4.0 min with a 170 L/min pump (ALR-CAL-001, C)                                                                                      |
| R9  | Data output               | Results saved as CSV with density, thrust, torque, speed, voltage and current at 10 Hz or more                                                                                               | Inspection of logged files                                                   | Met on paper: 20 Hz logging, amplifiers at 80 Hz (ALR-CAL-001, G)                                                                                 |
| R10 | Cost                      | Complete rig at or below USD 1,000                                                                                                                                                           | Costed bill of materials                                                     | Not met: estimated USD 4,920, over the value-engineering target by USD 3,920 (ALR-CAL-001, H)                                                     |
| R11 | Interlocks                | The motor can be powered only with the door latched and the arming key turned; the emergency stop removes motor and pump power; the bleed valve opens and vents the chamber on loss of power | Function check of each interlock before first power                          | Met on paper by design (ALR-BLD-001, sections 4 and 6); the door is also held shut by 24 kN of air load at the 5,000 m point                      |

R12

| ID | Requirement            | Target                                                                                                                                 | Verification (TRL 4 or later)                                         | Status at TRL 3                     |
|----|------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-------------------------------------|
|    | Over-vacuum protection | A relief valve stops the chamber going below 46.3 kPa absolute (55 kPa below atmosphere); the vessel is still designed for full vacuum | Relief valve set and checked against the gauge before first pump-down | Met on paper (ALR-CAL-001, A and B) |

## ■ Requirements not met or at risk

- **R6, at risk.** The closed loop of air inside the vessel adds a little resistance to the propeller's own flow, so the propeller is expected to give slightly more thrust in the chamber than in open air. Two simple models give 2.9 % and 5.3 % with the 63 % open baffle. Options and a recommendation are in `docs/REVIEW.md` for Amish to decide.
- **R10, not met.** The vessel parts, the welding, the motor supply and the safety controls take the estimate to USD 4,920. The budget is a value-engineering target, not a limit; options to bring the cost down are in `docs/REVIEW.md` and in the design decisions register.

## ■ Assumptions

- Matching air density is the main effect; the small differences in viscosity and speed of sound between a room-temperature chamber and real air at 5,000 m and -17.5 °C are second order for small propellers (to be checked). Blade Reynolds number falls by about 39 % at 5,000 m density (ALR-CAL-001, E), which the chamber reproduces because density, not pressure, is matched.
- A surplus or new-cut 813 mm steel pipe offcut and two tank heads can be bought; the out-of-roundness of the pipe is no more than 1 % of its diameter (to confirm when bought).
- The chamber effect is measured by side-by-side runs at ambient pressure and recorded with every published data set.
- Earlier physics estimate: hover time at 5,000 m and -20 °C is about 47 % of sea level; AltiRig will test the aerodynamic share of that estimate.

**Safety:** R7, R11 and R12 are safety requirements. They are met on paper only; no pump-down or motor run is made until the safety stops in the build plan (ALR-BLD-001, section 6) are passed and a qualified engineer has reviewed the vessel calculation.